

Newborough Warren

Weather & Water Watch

May 2026

Weather Summary

Metric	Newborough (ILLANF24)	RAF Valley	Typical May*
Total Rainfall	40.6 mm	34.0 mm	~52.4 mm
Average Highs		16.2°C	15.0°C
Average Lows		9.8°C	8.6°C
Sunshine		223.7 hrs	219.3 hrs
Air Frost		0 days	~0 days

**Based on 1991–2020 Met Office climate averages (RAF Valley). Local gauge: ILLANF24 (Weather Underground).*

May was the third consecutive drier-than-average month. RAF Valley recorded 34.0 mm, just 65% of the 52.4 mm long-term average. The local ILLANF24 gauge at Niwbwrch recorded 40.6 mm with a full 31 days of clean data.

The month started with three days of light rain before a nine-day dry spell set in from the 4th to the 12th — the longest dry run so far this spring. Mid-month brought scattered showers, with the wettest consecutive days on the 18th (6.3 mm) and 19th (5.1 mm). The most notable event came on the 27th: a sharp 12.2 mm downpour that delivered nearly a third of the month's total in a single day. The last four days were dry.

Temperatures were slightly above average: highs of 16.2°C (norm 15.0°C), lows of 9.8°C (norm 8.6°C). Sunshine was close to normal at 223.7 hours. No frost.

The hydrological year cumulative (October to May) stands at 681 mm, 114% of the 598 mm average. The buffer from the wet winter is shrinking — down from 126% at the end of March — but remains above normal.

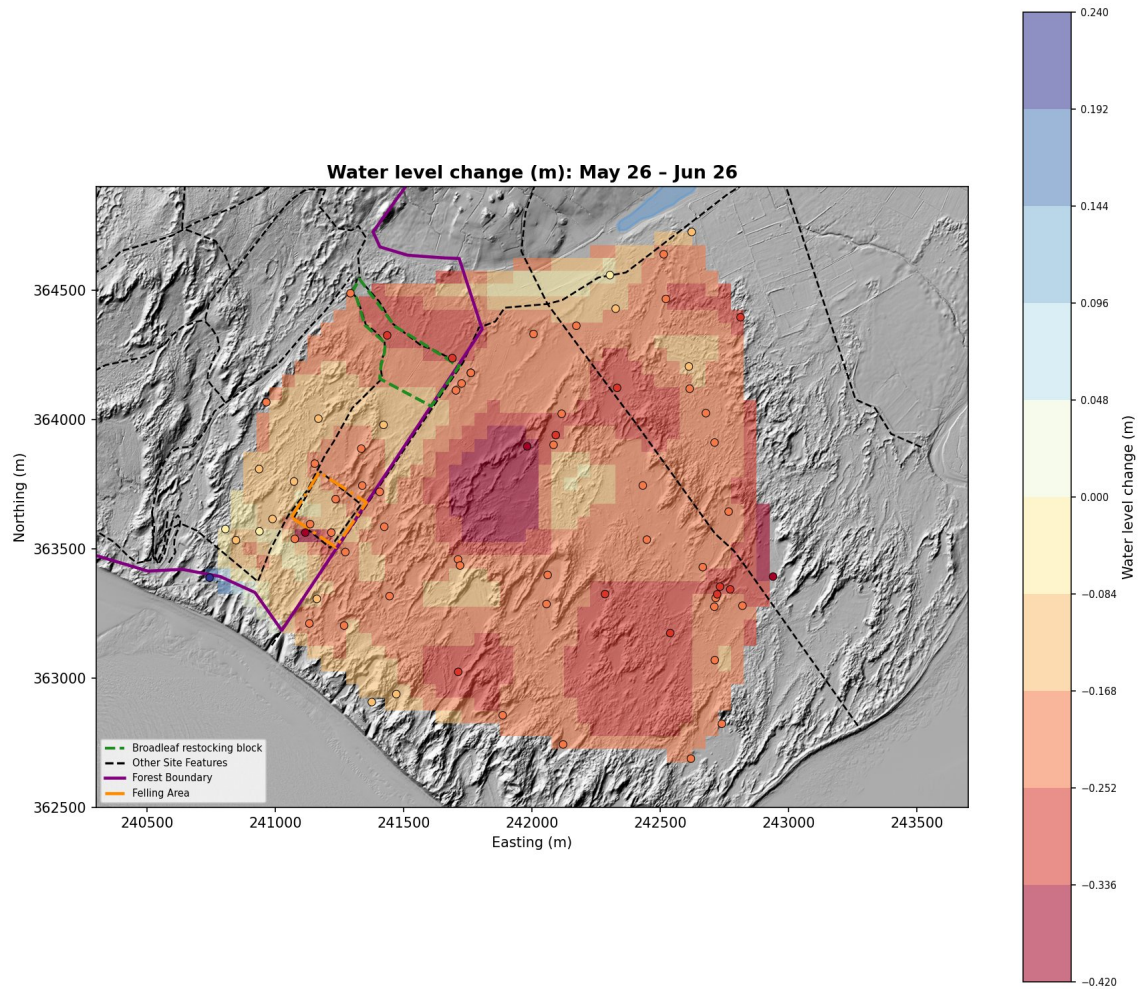
Local Water Levels

The well round on 4th June covered 74 dipwells. Of these, 73 declined and one rose. The mean change since the previous reading on 2nd May was -0.201 m — a slightly slower rate of decline than the -0.292 m recorded in the previous period.

The single riser was ceh3, which sits close to the shore and may be tidally influenced. Its $+0.220$ m rise is likely a correction from an anomalously low reading last month rather than a genuine recharge signal.

Month-on-Month Change: 2 May – 4 June 2026

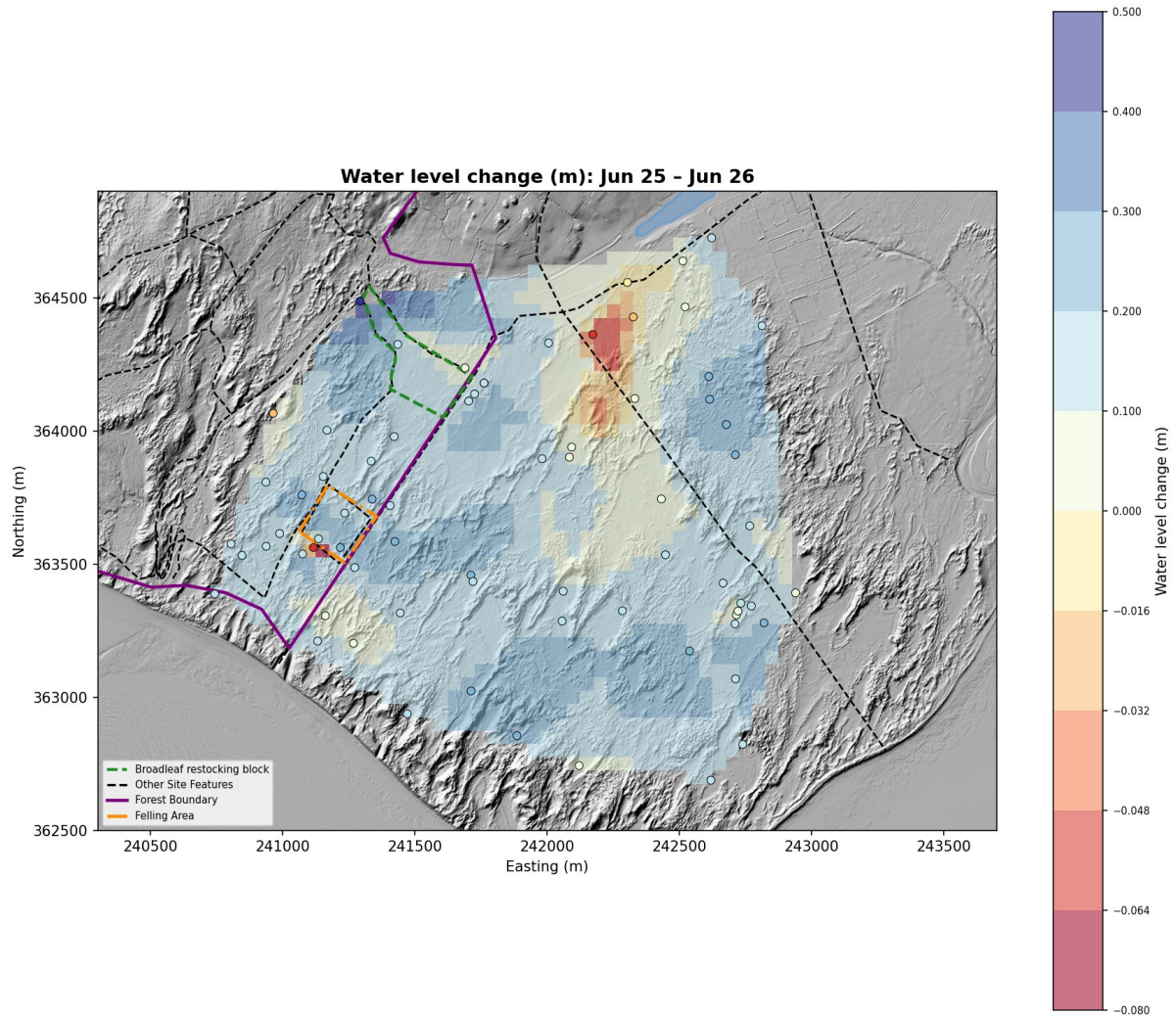
The largest declines were at ceh24 (-0.400 m), D15 and FE4 (both -0.390 m), and nw5 (-0.310 m). The smallest declines were at ceh17 (-0.050 m), nw9 and p2 (both -0.070 m).



Water level change (m) from 02 May 2026 to 04 Jun 2026. *ceh3* (far west) is the sole riser.

Year-on-Year: June 2025 vs June 2026

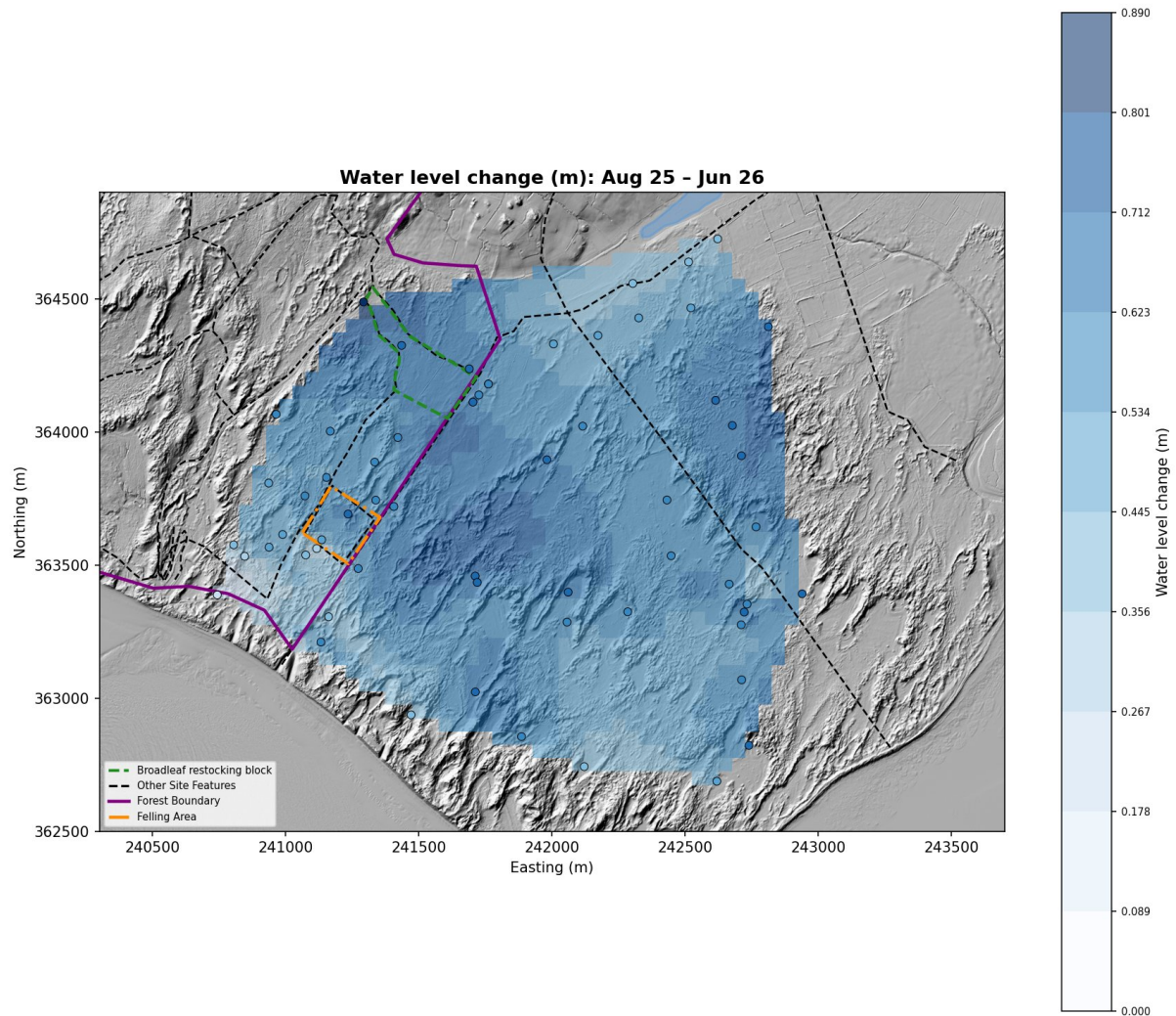
The year-on-year advantage continues to narrow. Of 73 comparable wells, 66 remain higher than this time last year, but 4 have now dropped below their June 2025 levels. The mean year-on-year difference is +0.137 m — down from +0.247 m last month and +0.349 m in March.



Year-on-year water level change from 03 Jun 2025 to 04 Jun 2026.

Rebound from Summer Low: August 2025 – June 2026

All 61 comparable wells remain above their August 2025 summer low, with a mean cumulative rise of +0.556 m (range +0.190 to +0.840 m). The largest cumulative gain is at ceh14 (+0.840 m).



Cumulative water level change since 31 Aug 2025.

Mean Spring Water Level 2026

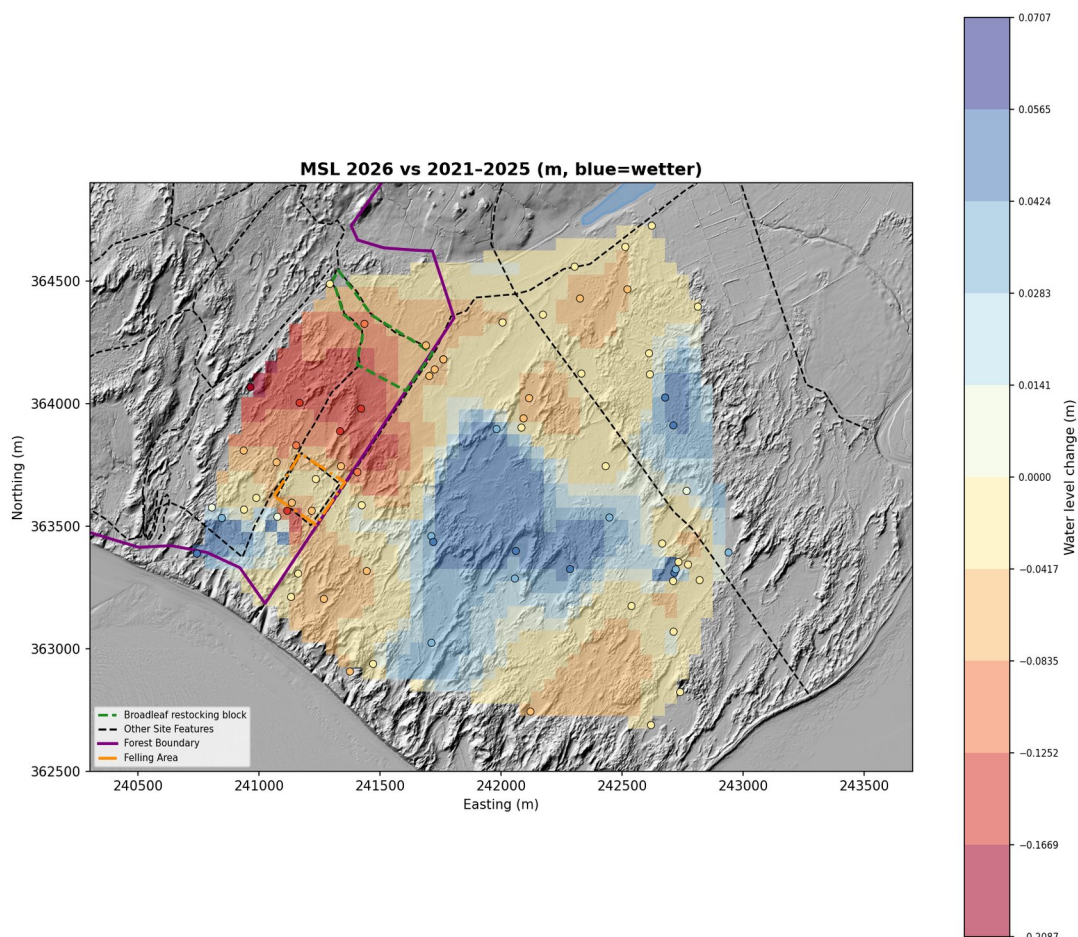
The Mean Spring Water Level (MSL) is the average depth to water table below ground surface across the March, April, and May readings. It is a standard ecohydrological index for dune slack habitat assessment. Lower values indicate wetter conditions. This year's MSL is final with all three spring months recorded, and is compared to the 2021–2025 five-year reference.

The 2026 MSL is +0.027 m deeper (drier) than the 5-year benchmark. Of 73 wells, 48 are drier than their benchmark, 15 are wetter, and 10 are essentially unchanged. Despite the very wet winter, the dry April and May pulled the spring mean back towards average. The result is a spring that was, on balance, close to — but fractionally below — the recent norm.

The wells that finished wetter than their benchmark are concentrated in the central and eastern dune slacks: ceh40, ceh41, D8, D9, and ceh9. These are the wells where the winter recharge pulse penetrated deepest and persisted longest into spring. The wells furthest below their benchmark are in the western forest block: ceh13 (+0.189 m drier), ceh2 (+0.161 m), Ceh32 and FE4 (both +0.127 m). These are the wells most affected by canopy interception and transpiration from the Corsican pine plantation.

MSL 2026 vs 5-Year Reference

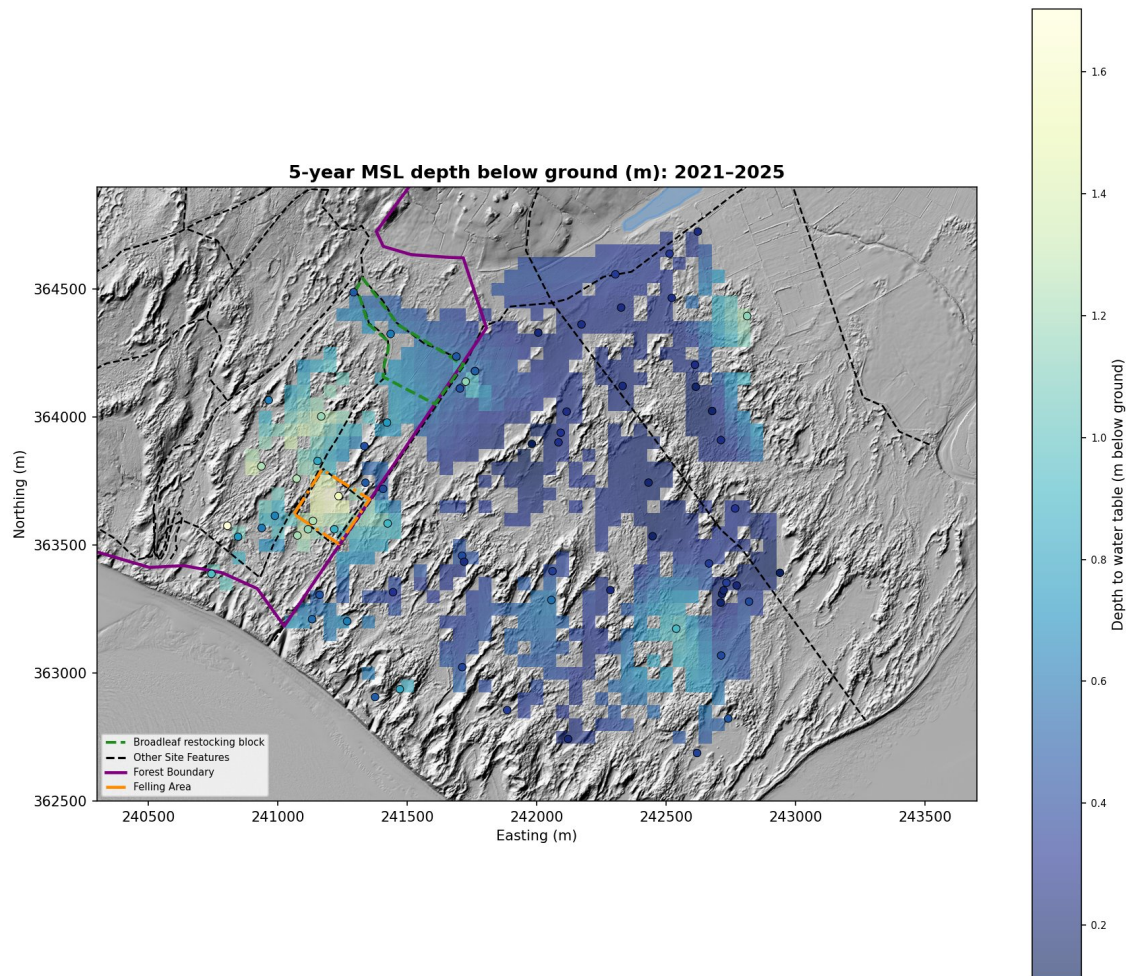
The MSL difference map shows the spatial pattern. Blue areas had a wetter spring than the benchmark; red areas were drier. The central slacks around D15, ceh24, and the T41 cluster are blue — these benefited from the wet winter. The western and northern forest block is red, reflecting the ongoing transpiration demand from the plantation canopy.



MSL 2026 vs 2021–2025 reference. Blue = wetter than benchmark, red = drier.

5-Year Reference MSL (2021–2025)

The reference map shows the spatial pattern of spring water table depth across the reserve. The deepest water tables (yellow tones) are in the western forest block and around ceh17 and ceh30, where the plantation canopy draws water down. The shallowest (blue/purple tones) are in the central and southeastern slacks — the T41 cluster, D15, D8, and D9 — where the water table is typically within 0.2 m of the surface during spring. The dune ridges are masked out since MSL is only meaningful in the low-lying slacks.

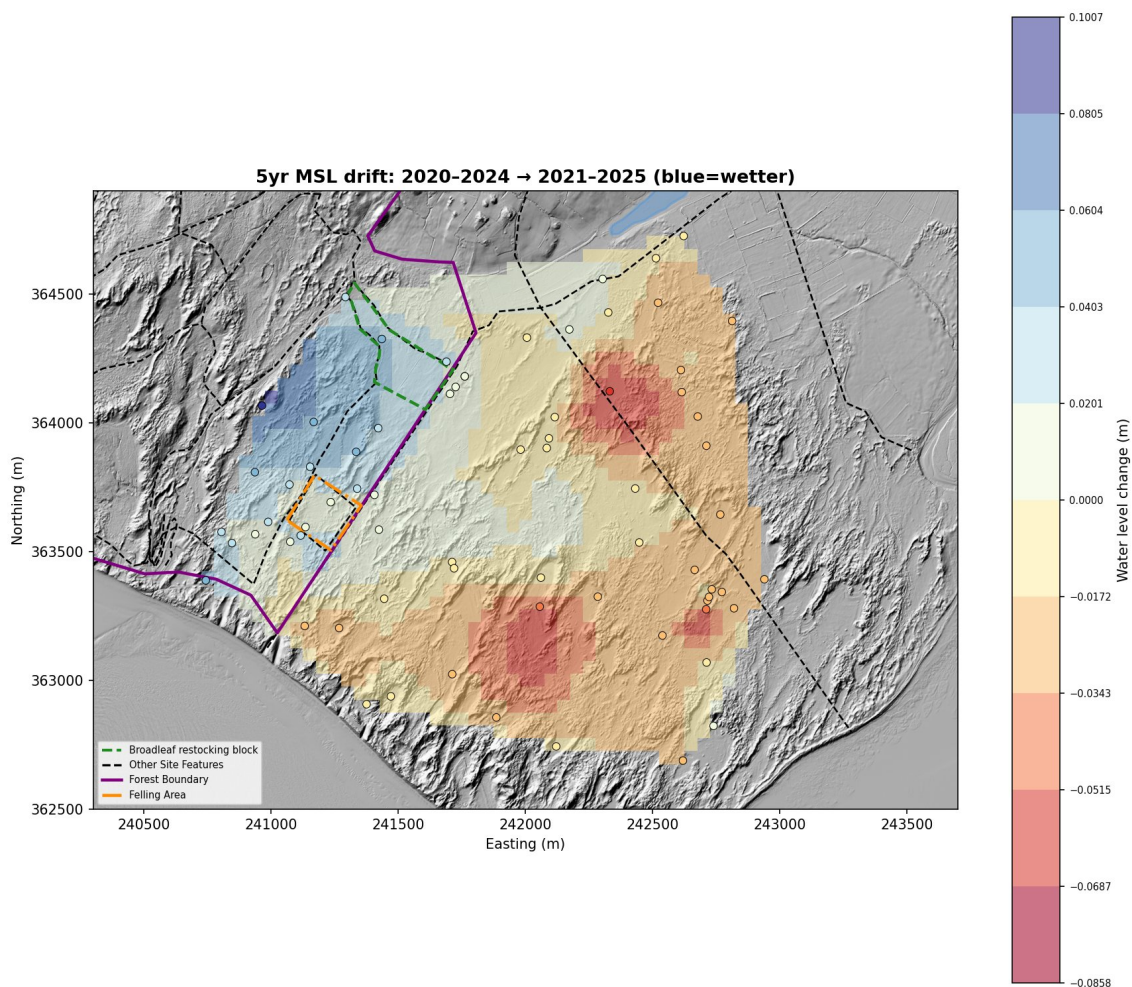


Five-year MSL depth below ground (m), 2021–2025. Dune ridges masked. Blue/purple = shallow (wet), yellow = deep (dry).

5-Year MSL Trend

The drift map shows how the 5-year MSL benchmark itself has shifted compared to the previous year's benchmark (2020–2024 vs 2021–2025). Blue = benchmark getting wetter; red = getting drier.

The western forest block shows the benchmark rising (getting wetter) — consistent with reduced transpiration following clearfell operations. The central and southeastern dune slacks show the benchmark drifting drier, reflecting the dry springs of 2022 and 2025 entering the rolling average while the wetter 2020 drops out. This pattern is ecologically significant: it suggests improving conditions in the clearfell area and a declining trend in the eastern dunes that warrants continued monitoring.



Change in 5-year MSL reference: 2020–2024 vs 2021–2025. Blue = benchmark getting wetter, red = drier.

Well Measurement Summary

Risers and Largest Declines (2 May – 4 June)

Smallest change	Change (m)		Largest decline	Change (m)
ceh3	+0.220		ceh10	-0.300
ceh17	-0.050		ceh28	-0.300
nw9	-0.070		nw5	-0.310
p2	-0.070		D15	-0.390
ceh22	-0.100		ceh24	-0.400

74 wells measured. 73 falling, 1 rising. Mean: -0.201 m. Range: -0.400 to +0.220 m.

Outlook

The recession will continue into June. With the canopy at full leaf and the longest days approaching, evapotranspiration rates will remain high. The year-on-year surplus is now substantially reduced (+0.137 m, down from +0.349 m in March) and could be largely spent by late summer.

The MSL analysis shows that despite the exceptional winter recharge, Spring 2026 ended close to the 5-year benchmark. The ecological implication is that one wet winter is not enough to shift the spring average significantly — the dry months that follow quickly erode the gains. The drift map provides an encouraging signal from the clearfell area, where the long-term benchmark is trending wetter.